

SMU Lyle School of Engineering

Reinforcement Learning

Eligibility Traces

Sutton/Barto* Chapter 12

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018



Online Material

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Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- Part I: Tabular Methods
  - Dynamic programming
  - Monte Carlo methods
  - Temporal-difference learning
  - Multi-step bootstrapping
  - Planning and learning with tabular methods
- **Part II: Approximate Solution Methods**
  - Prediction and Control using Approximation
  - **Eligibility Traces**
  - Policy Gradient Methods
- Part III: Modern RL Methods
  - Deep Reinforcement Learning
  - Current Applications

Can also be used instead of n-step methods for tabular methods!

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\boldsymbol{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\boldsymbol{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X, i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\text{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or v_*

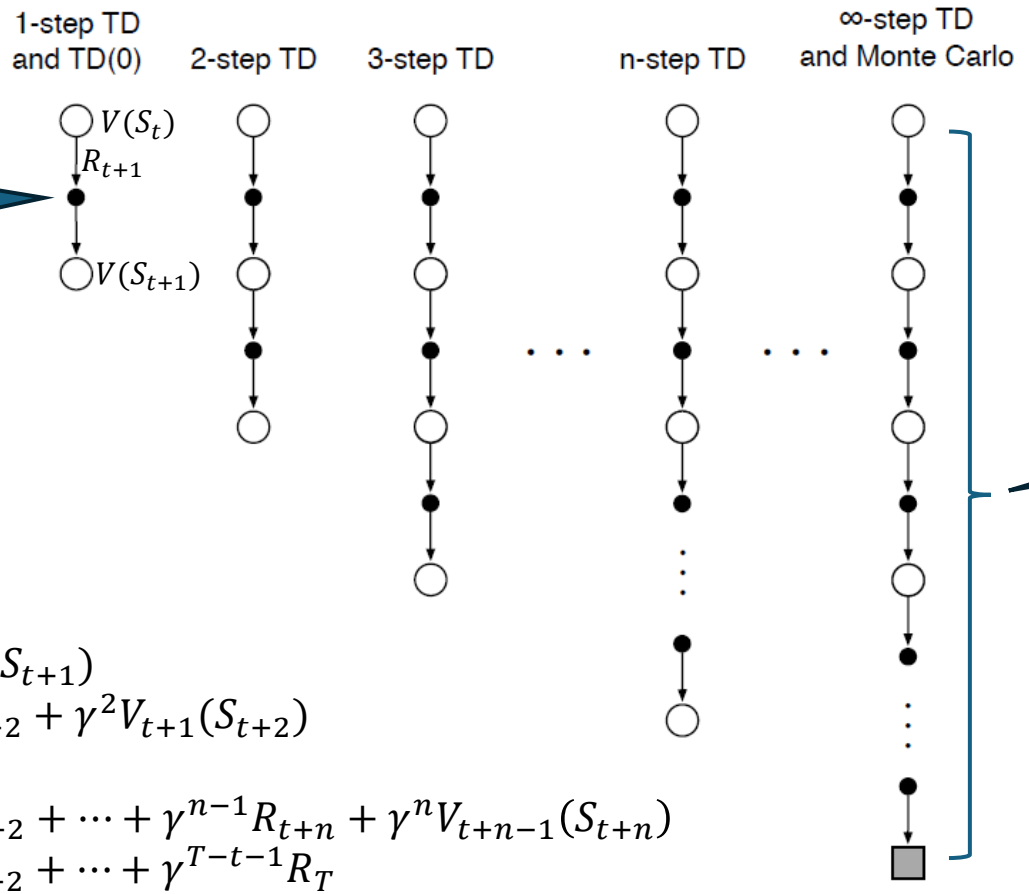
MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r \mid s, a)$	probability of transition to state s' and receiving reward r, from state s taking action a.
$p(s' \mid s, a)$	probability of transition to state s' fom state s taking action a.
$r(s, a)$	expected immediate reward from state s after action a.
$r(s, a, s')$	expected immediate reward from state s to s' with action a.
$\pi(a \mid s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

The λ-Return

Performing updates with compound return targets

Remember Chapter 7: n-step TD Returns



Update target is the reward R_{t+1} + bootstrap the remaining return by the value of the next state $V(S_{t+1})$.

Update with the whole sequence of observed rewards → no bootstrapping.

Return

1-step TD: $G_{t:t+1} = R_{t+1} + \gamma V_t(S_{t+1})$

2-step TD: $G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V_{t+1}(S_{t+2})$

…

n-step TD: $G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n V_{t+n-1}(S_{t+n})$

MC: $G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$

Update: $V(S_t) \leftarrow V(S_t) + \alpha[G_{t:t+n} - V(S_t)]$

The update target $G_{t:t+n}$ is only available with a n-steps delay!

Notation

$G_{t:t+n}$ … the return is calculated from the rewards up to $t+n$ and the rest is bootstrapped using the value function estimate.

Valid n-step Update Targets and Compound Updates

- n-step return from Chapter 7 with approximation:

Episode termination

$$G_{t:t+n} = \underbrace{R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n}}_{n \text{ next rewards}} + \underbrace{\gamma^n \hat{v}(S_{t+n}, \boldsymbol{w}_{t+n-1})}_{\text{approx. value}}, \qquad 0 \le t \le T - n$$

- Any n-step return $G_{t:t+n}$ has the **error reduction property** (TD policy evaluation is γ-contracting): It is a **valid update target** for tabular learning and function approximation.

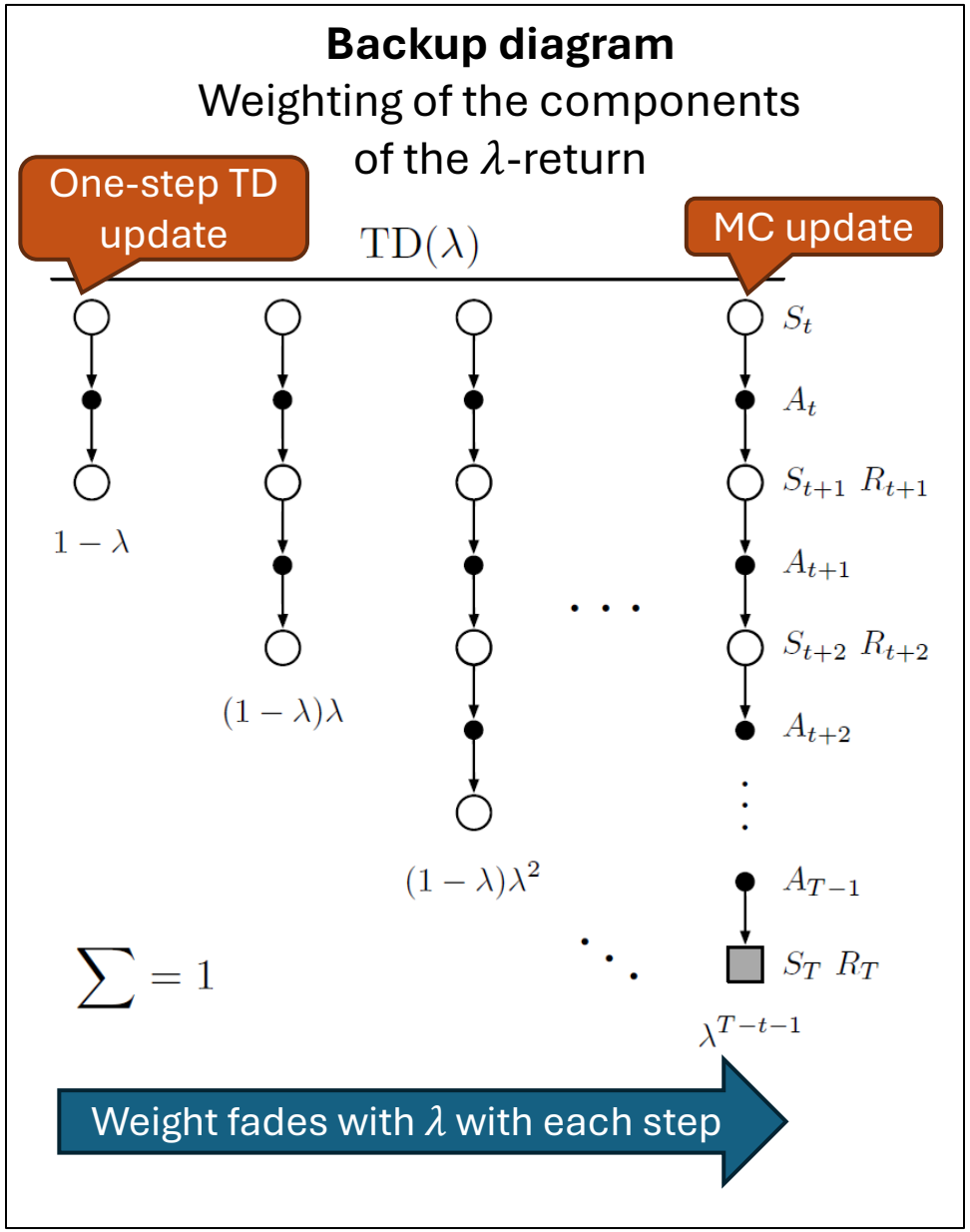
- **Compound updates:**
  - Any average of different n-step returns is also valid target.
  - Updating with an average of n-step returns reduces the error.
    E.g., average 2-step and 4-step returns: $\frac{1}{2} G_{t:t+2} + \frac{1}{2} G_{t:t+4}$

TD(λ) and the λ-Return

- TD(λ) uses a compound return called the ***λ*-return** as the update target:

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} G_{t:t+n}$$

- Special cases:
  - **TD(0)** with $\lambda = 0$ uses only first component. One-step TD update.
  - **TD(1)** with $\lambda = 1$ uses only the last component. MC update.

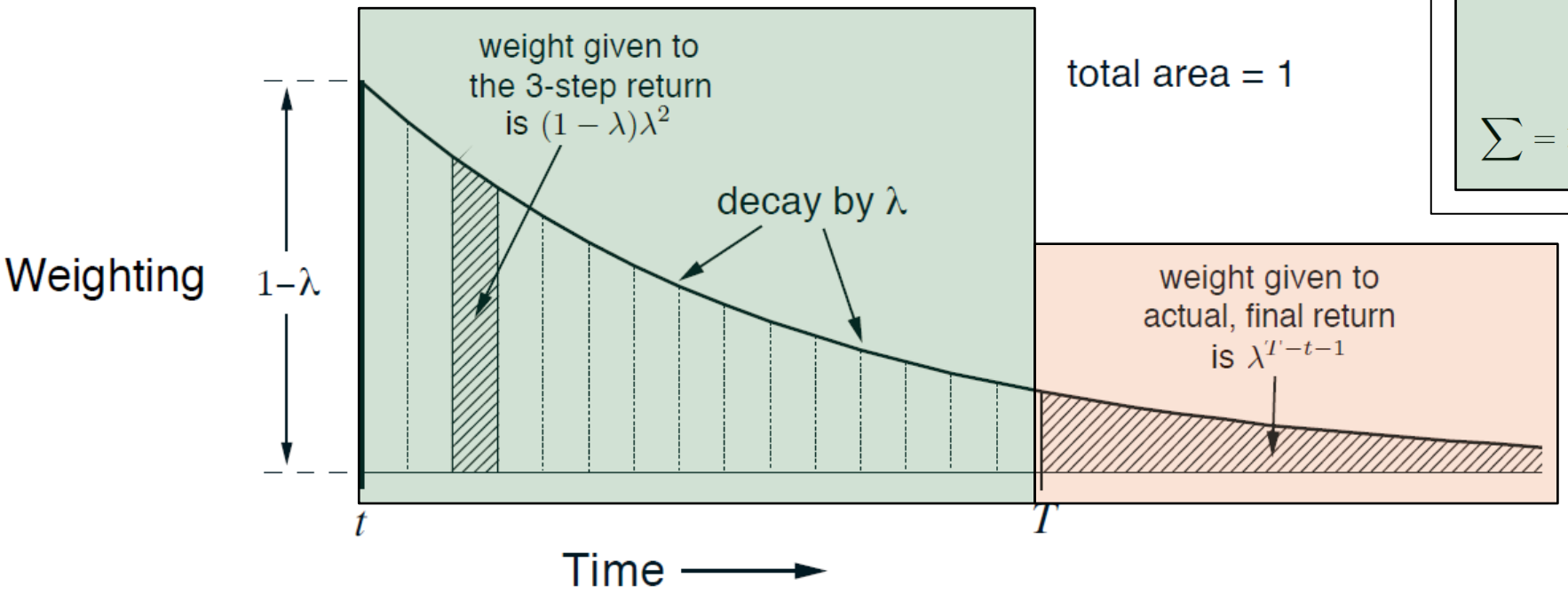


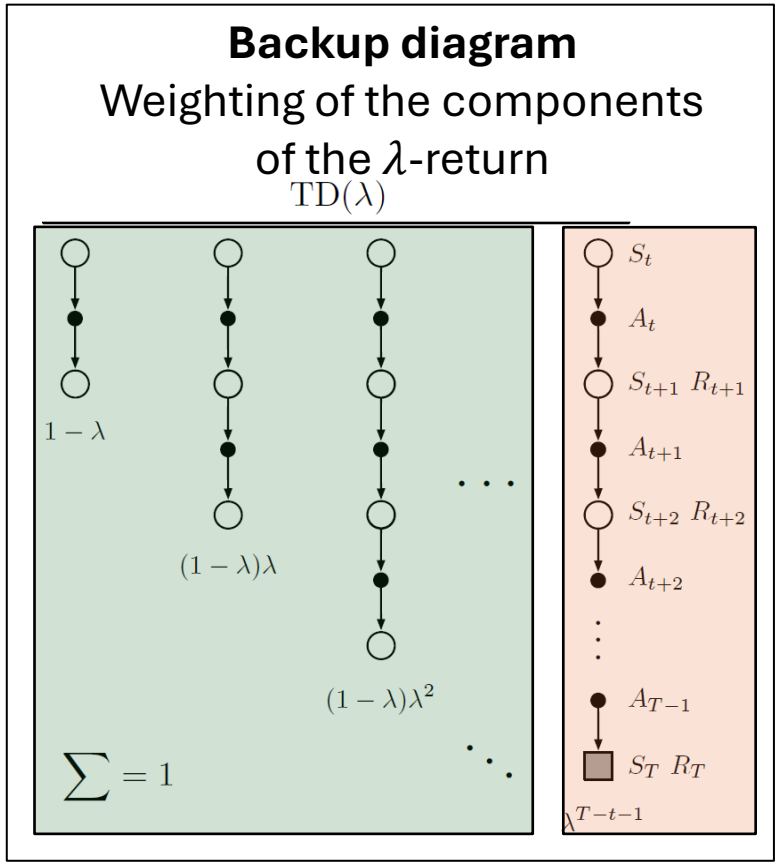
Exponential Weighting Scheme of the λ-Return

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{T-t-1} \lambda^{n-1} G_{t:t+n} + \lambda^{T-t-1} G_t$$

n-step returns

MC return



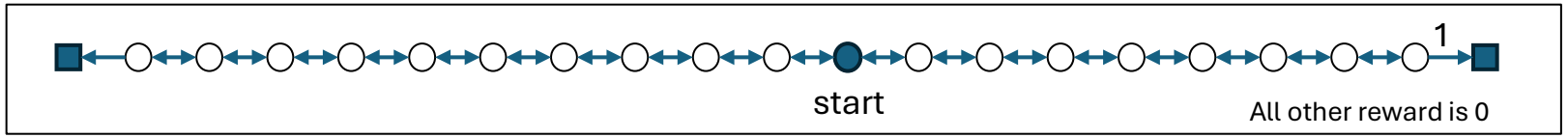


The Off-Line λ-Return Algorithm

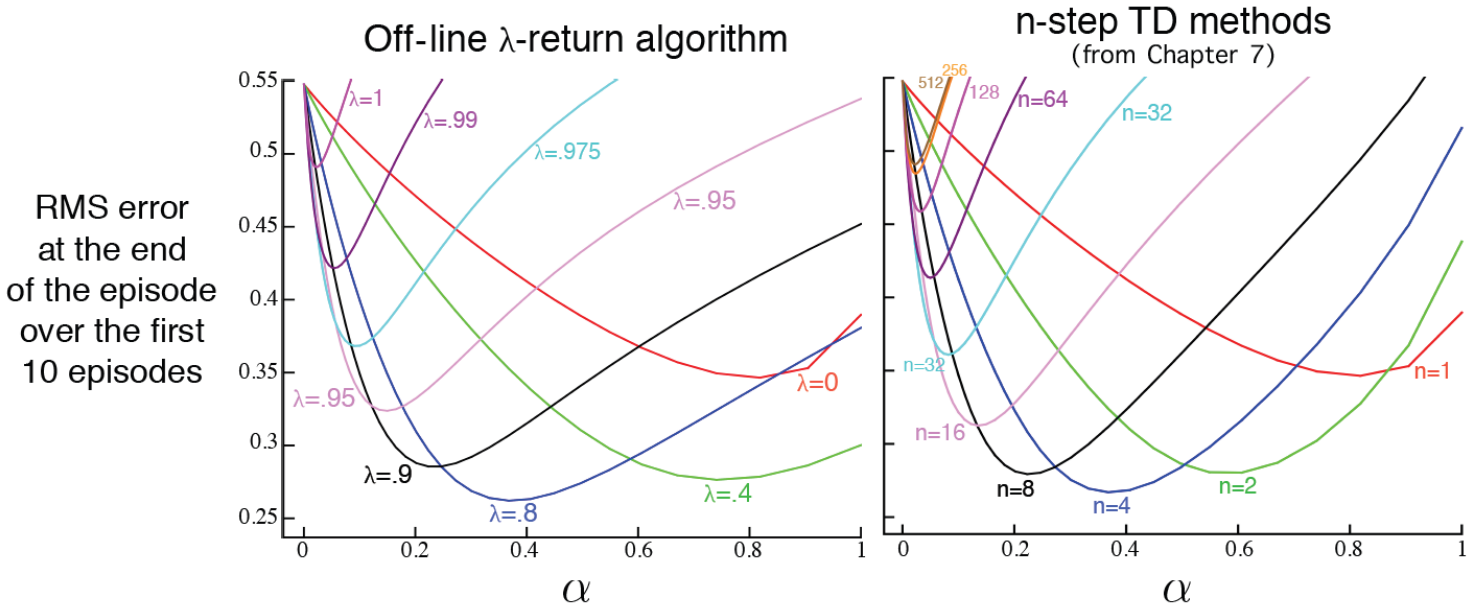
- A simplistic algorithm is to use the λ-return as the update target for **semi-gradient descent with approximation**.
- **Update**: Use $U_t = G_t^\lambda$ as the target

$$\boldsymbol{w}_{t+1} = \boldsymbol{w}_t + \alpha\left[G_t^\lambda - \hat{v}(S_t, \boldsymbol{w}_t)\right]\nabla\hat{v}(S_t, \boldsymbol{w}_t), \qquad t = 0, \ldots, T-1$$

Example: 19-state random walk value estimation task.

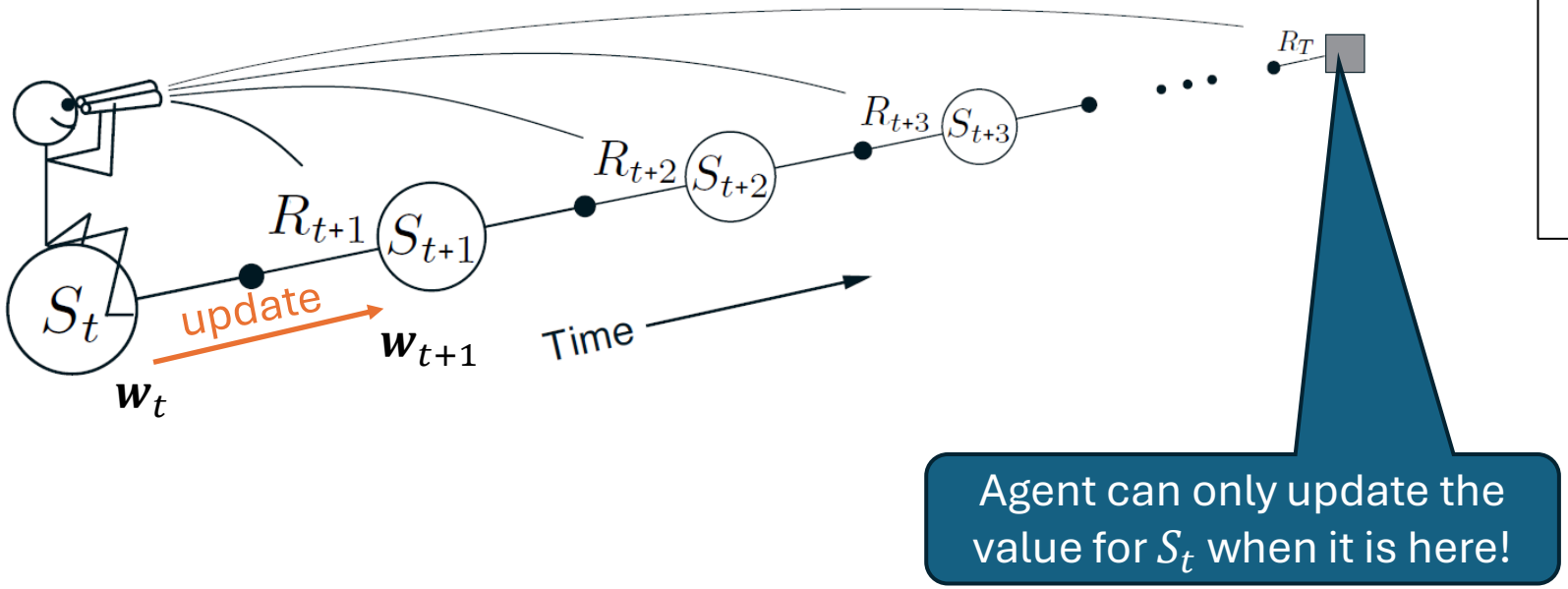


λ-return algorithm mimics n-step methods.



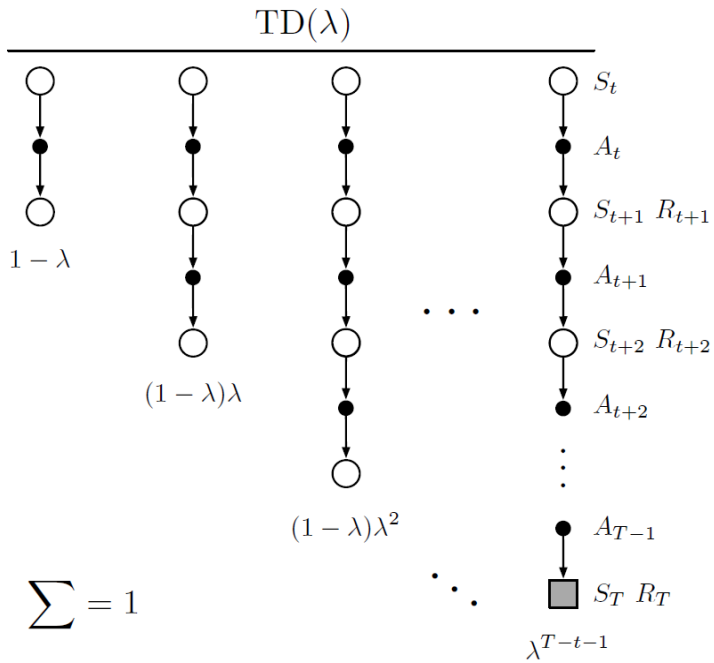
The Off-Line λ-Return Algorithm: The Forward View Issue

- Uses the **"forward view."**
- Updating S_t requires the calculation of all $G_{t:t+1}$ to G_t and thus the whole episode.
- The **update is delayed** till the whole episode is observed (like for MC).



Backup diagram

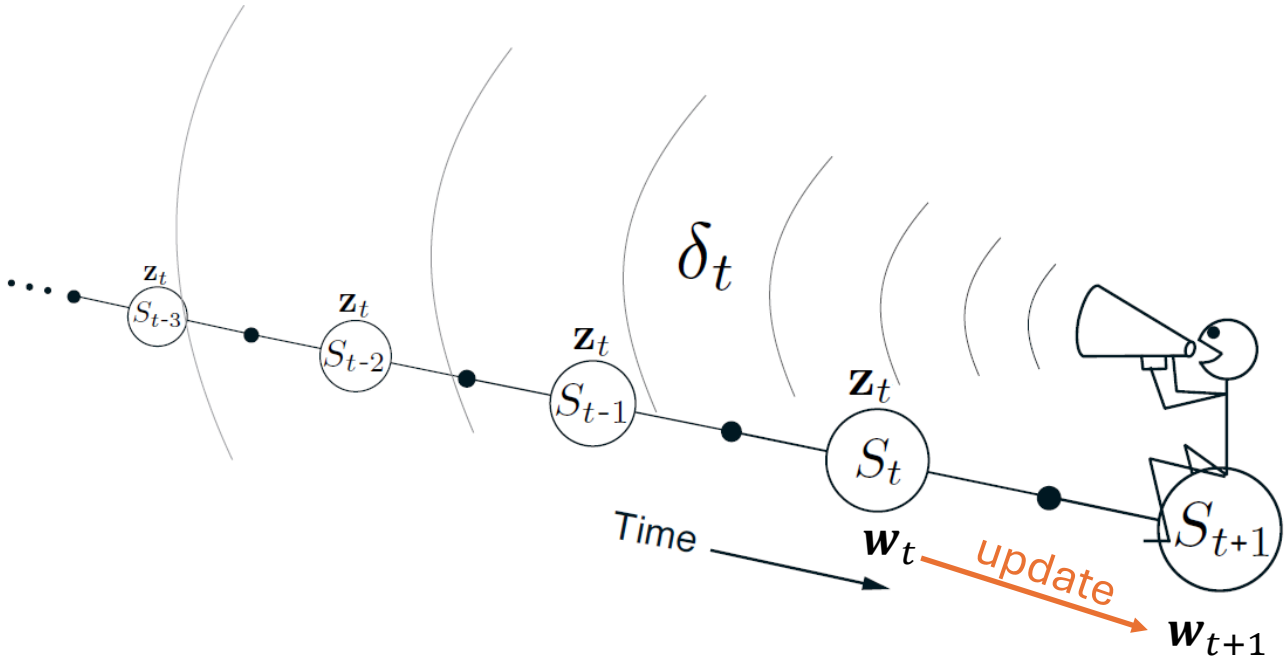
Weighting of the components of the λ-return



The Backward View

Using eligibility traces

The Backward View



δ_t is the TD-error

z_t remembers how much to blame state for δ_t.

If we can **update the value function for all previous states** depending on the current observed TD-error:

- Approximate the behavior of the forward-looking Off-Line λ-Return Algorithm.
- Online update of $\boldsymbol{w}$ at every time step.
- Computations are equally spread over time (not just the end of the episode).
- Works for continuing problems with no episode structure.

Eligibility Traces

- The approximation weights

$$\boldsymbol{w}_t \in \mathbb{R}^d$$

  represent what we have learned about the value function so far = **long-term memory**.

- **Eligibility trace:** a vector

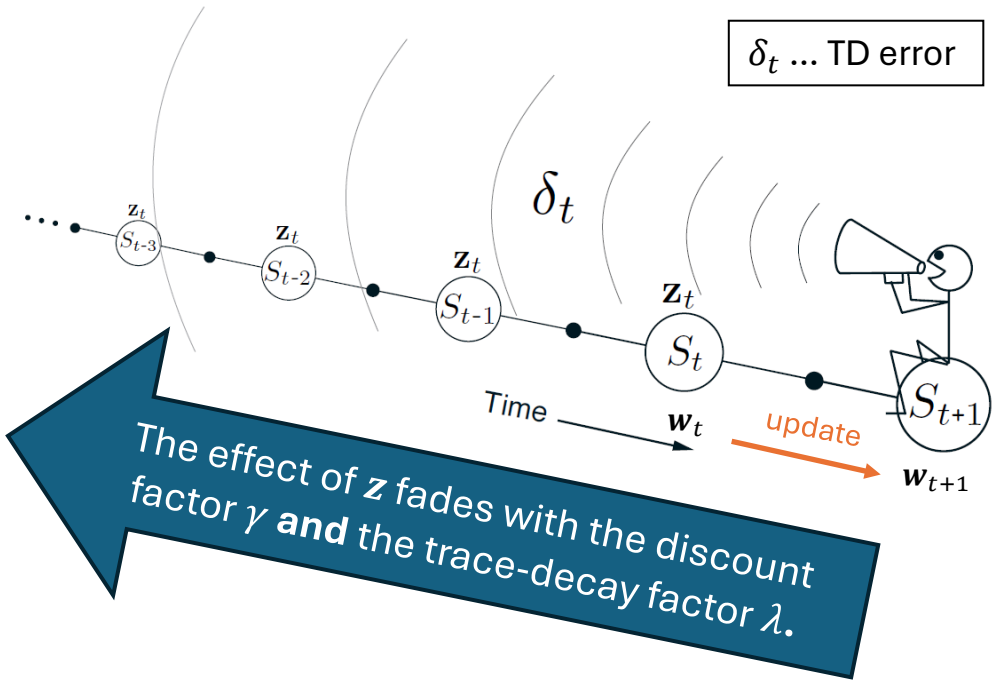
$$\boldsymbol{z}_t \in \mathbb{R}^d$$

- **Update**:

$$\boldsymbol{z}_t = \gamma\lambda\boldsymbol{z}_{t-1} + \nabla\hat{v}(S_t, \boldsymbol{w}_t)$$
$$\boldsymbol{w}_{t+1} = \boldsymbol{w}_t + \alpha\delta_t\boldsymbol{z}_t$$

- Each component in $\boldsymbol{z}_t$ keeps track of how much each component in $\boldsymbol{w}_t$ contributed to recent state valuations in the form of a **weighted sum of past gradients**.

- **Interpretation:**
  - $\boldsymbol{z}_t$ represents a **short-term memory** for credit assignment.
  - $\boldsymbol{z}_t$ shows the **eligibility** of each component of $\boldsymbol{w}_t$ to undergo learning changes when a reinforcement event happens.
  - The update perfectly matches the forward view update with G_t^λ!



Semi-gradient TD(λ) for Estimation

Semi-gradient TD(λ) for estimating $\hat{v} \approx v_\pi$

Input: the policy π to be evaluated
Input: a differentiable function $\hat{v} : \mathcal{S}^+ \times \mathbb{R}^d \to \mathbb{R}$ such that $\hat{v}(\text{terminal},\cdot) = 0$
Algorithm parameters: step size $\alpha > 0$, trace decay rate $\lambda \in [0, 1]$
Initialize value-function weights $\mathbf{w}$ arbitrarily (e.g., $\mathbf{w} = \mathbf{0}$)

Loop for each episode:
  Initialize S
  $\mathbf{z} \leftarrow \mathbf{0}$ (a d-dimensional vector)
  Loop for each step of episode:
  | Choose $A \sim \pi(\cdot|S)$
  | Take action A, observe R, S'
  | $\mathbf{z} \leftarrow \gamma\lambda\mathbf{z} + \nabla\hat{v}(S,\mathbf{w})$
  | $\delta \leftarrow R + \gamma\hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$
  | $\mathbf{w} \leftarrow \mathbf{w} + \alpha\delta\mathbf{z}$
  | $S \leftarrow S'$
  until S' is terminal

True Online TD(λ) for Estimation

States are represented by feature vectors $\boldsymbol{x}$
Replace $\hat{v}(S, \boldsymbol{w})$ with $\mathrm{V} = \boldsymbol{w}^\top \boldsymbol{x}$

Semi-gradient TD(λ) for estimati

Input: the policy π to be evaluated
Input: a differentiable function $\hat{v} : \mathcal{S}^+$
Algorithm parameters: step size $\alpha > 0$
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Loop for each episode:
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  | $\mathbf{z} \leftarrow \gamma\lambda\mathbf{z} + \nabla\hat{v}(S,\mathbf{w})$
  | $\delta \leftarrow R + \gamma\hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$
  | $\mathbf{w} \leftarrow \mathbf{w} + \alpha\delta\mathbf{z}$
  | $S \leftarrow S'$
  until S' is terminal

True online TD(λ) for estimating $\mathbf{w}^\top\mathbf{x} \approx v_\pi$

Input: the policy π to be evaluated
Input: a feature function $\mathbf{x} : \mathcal{S}^+ \to \mathbb{R}^d$ such that $\mathbf{x}(terminal, \cdot) = \mathbf{0}$
Algorithm parameters: step size $\alpha > 0$, trace decay rate $\lambda \in [0, 1]$
Initialize value-function weights $\mathbf{w} \in \mathbb{R}^d$ (e.g., $\mathbf{w} = \mathbf{0}$)

Loop for each episode:
  Initialize state and obtain initial feature vector $\mathbf{x}$
  $\mathbf{z} \leftarrow \mathbf{0}$ (a d-dimensional vector)
  $V_{old} \leftarrow 0$ (a temporary scalar variable)
  Loop for each step of episode:
  | Choose $A \sim \pi$
  | Take action A, observe R, $\mathbf{x}'$ (feature vector of the next state)
  | $V \leftarrow \mathbf{w}^\top\mathbf{x}$
  | $V' \leftarrow \mathbf{w}^\top\mathbf{x}'$
  | $\delta \leftarrow R + \gamma V' - V$
  | $\mathbf{z} \leftarrow \gamma\lambda\mathbf{z} + \left(1 - \alpha\gamma\lambda\mathbf{z}^\top\mathbf{x}\right)\mathbf{x}$
  | $\mathbf{w} \leftarrow \mathbf{w} + \alpha(\delta + V - V_{old})\mathbf{z} - \alpha(V - V_{old})\mathbf{x}$
  | $V_{old} \leftarrow V'$
  | $\mathbf{x} \leftarrow \mathbf{x}'$
  until $\mathbf{x}' = \mathbf{0}$ (signaling arrival at a terminal state)

The additional Vs make sure that the update exactly matches the forward view.

Control with Eligibility Traces

Sarsa(λ)

Sarsa(λ)

True online TD(λ) for estimating $\mathbf{w}^\top\mathbf{x} \approx v_\pi$

Input: the policy π to be evaluated
Input: a feature function $\mathbf{x} : \mathcal{S}^+ \to \mathbb{R}^d$ such that $\mathbf{x}(terminal,$
Algorithm parameters: step size $\alpha > 0$, trace decay rate $\lambda \in [$
Initialize value-function weights $\mathbf{w} \in \mathbb{R}^d$ (e.g., $\mathbf{w} = \mathbf{0}$)

Loop for each episode:
  Initialize state and obtain initial feature vector $\mathbf{x}$
  $\mathbf{z} \leftarrow \mathbf{0}$ (a d-di
  $V_{old} \leftarrow 0$ (a tem
  Loop for each step of episode:
  | Choose $A \sim \pi$
  | Take action A, observe R, $\mathbf{x}'$ (feature vector of the
  | $V \leftarrow \mathbf{w}^\top\mathbf{x}$
  | $V' \leftarrow \mathbf{w}^\top\mathbf{x}'$
  | $\delta \leftarrow R + \gamma V' - V$
  | $\mathbf{z} \leftarrow \gamma\lambda\mathbf{z} + \left(1 - \alpha\gamma\lambda\mathbf{z}^\top\mathbf{x}\right)\mathbf{x}$
  | $\mathbf{w} \leftarrow \mathbf{w} + \alpha(\delta + V - V_{old})\mathbf{z} - \alpha(V - V_{old})\mathbf{x}$
  | $V_{old} \leftarrow V'$
  | $\mathbf{x} \leftarrow \mathbf{x}'$
  until $\mathbf{x}' = \mathbf{0}$ (signaling arrival at a te ate)

- Estimate action values $\hat{q}(s, a, \boldsymbol{w})$. V becomes Q.
- $\boldsymbol{x}$ is now a state+action feature

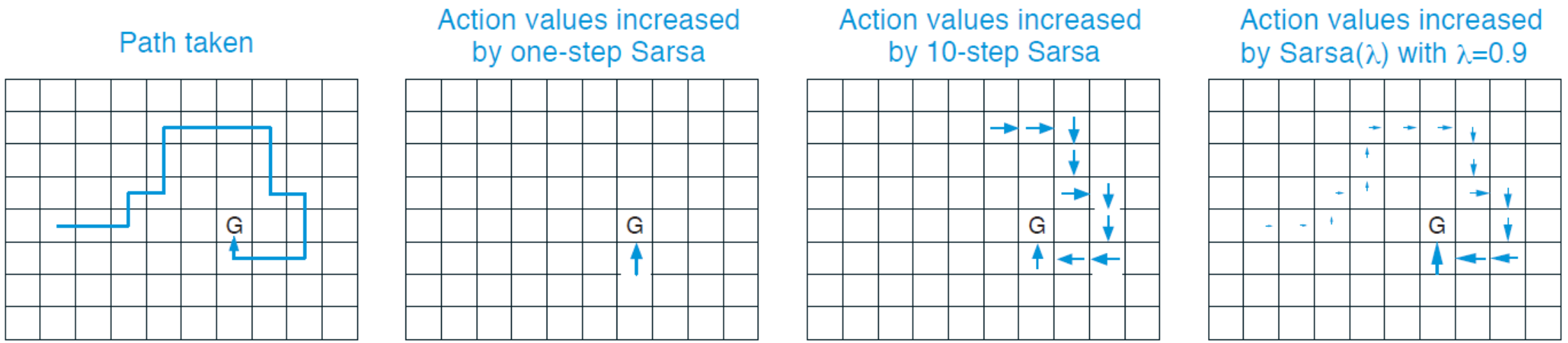
True online Sarsa(λ) for estimating $\mathbf{w}^\top\mathbf{x} \approx$ ~~q_π or~~ q_*

Input: a feature function $\mathbf{x} : \mathcal{S}^+ \times \mathcal{A} \to \mathbb{R}^d$ such that $\mathbf{x}(terminal, \cdot) = \mathbf{0}$
~~Input: a policy π (if estimating q_π)~~
Algorithm parameters: step size $\alpha > 0$, trace decay rate $\lambda \in [0, 1]$, small $\varepsilon > 0$
Initialize: $\mathbf{w} \in \mathbb{R}^d$ (e.g., $\mathbf{w} = \mathbf{0}$)

Loop for each episode:
  Initialize S
  Choose $A \sim \pi(\cdot|S)$ or ε-greedy according to $\hat{q}(S, \cdot, \mathbf{w})$
  $\mathbf{x} \leftarrow \mathbf{x}(S, A)$
  $\mathbf{z} \leftarrow \mathbf{0}$
  $Q_{old} \leftarrow 0$
  Loop for each step of episode:
  | Take action A, observe R, S'
  | Choose $A' \sim \pi(\cdot|S')$ or ε-greedy according to $\hat{q}(S', \cdot, \mathbf{w})$
  | $\mathbf{x}' \leftarrow \mathbf{x}(S', A')$
  | $Q \leftarrow \mathbf{w}^\top\mathbf{x}$
  | $Q' \leftarrow \mathbf{w}^\top\mathbf{x}'$
  | $\delta \leftarrow R + \gamma Q' - Q$
  | $\mathbf{z} \leftarrow \gamma\lambda\mathbf{z} + \left(1 - \alpha\gamma\lambda\mathbf{z}^\top\mathbf{x}\right)\mathbf{x}$
  | $\mathbf{w} \leftarrow \mathbf{w} + \alpha(\delta + Q - Q_{old})\mathbf{z} - \alpha(Q - Q_{old})\mathbf{x}$
  | $Q_{old} \leftarrow Q'$
  | $\mathbf{x} \leftarrow \mathbf{x}'$
  | $A \leftarrow A'$
  until S' is terminal

On-policy learning! Needs to keep exploring

Example: Traces in a Gridworld



- The first panel shows the path taken by an agent in a single episode.
- The initial estimated values were zero, and all rewards were zero except for a positive reward at the goal location marked by G.
- The arrows in the other panels show, for various algorithms, which action-values would be increased, and by how much, upon reaching the goal.
- Sarsa(λ) updates more values and also updates values closer to the goal more. This leads to **faster and more robust learning.**

Code example:
Semi-gradient Sarsa(λ):
Control for the L-maze grid world.

See Code on https://michael.hahsler.net/Introduction_to_Reinforcement_Learning/

Implementation Issues

- **Off-policy learning**: Use importance sampling (see Chapter 7). Note: Semi-gradient methods do not guarantee **stability** and may need extensions to improve stability.
- **Sparse traces**: Most values of $\boldsymbol{z}_t$ are typically close to 0. Many algorithms set very small values to zero and save computation.

- **How to chose $\boldsymbol{\lambda}$**: There is no clear optimal value. It depends on:
  - Lower for highly stochastic environments (sample variance)
  - Higher for sparse rewards
  - Inverse relationship with learning rate α to avoid divergence.

  Practitioners often start with lambda equals 0.9 and then lower it if behavior is unstable.

Issues With Non-Linear Approximation and Deep Reinforcement Learning

- **Non-linear approximation:** Gradient changes with $\boldsymbol{w}$, which breaks compound updates used by traces.
- **Deep reinforcement learning (DRL) uses experience replay:** Traces are sequential and do not work with a replay buffer.
- **DRL is typically off-policy:** off-policy corrections (like importance sampling) over deep networks cause the variance to explode exponentially, leading to divergence.
- **In practice**: Use n-step methods for DRL.





What You Should Know

- Eligibility traces **generalize n-step methods**: Move between MC and one-step TD.
- Eligibility traces are very general, often learn faster, and are **computationally efficient.**
- Since eligibility traces can move close to MC behavior, they should be used if
  - the problem is suspected to be **partially non-Markov** or
  - the **rewards** are very **delayed**.
- Eligibility traces **can also be used to extend Q-learning**. This leads to algorithms like Watkins's $Q(\lambda)$ and Tree-Backup(λ).
- **Non-linear approximation and DRL** are difficult. n-step methods are typically used.